MY RESUME

Zilong Jia

Research interests：
Multimodal Large Language Models, 3D, Multimodal Generation.

Birthday： 1999.5.12
Tel： +86 15147255045
E-mail： jiazilong9@gmail.com
Location： Beijing, China

EDUCATION

2018.09-2022.06	BUPT	School of Automation
2022.09-2025.06	BUPT	School of Artificial Intelligence

ACADEMIC ACHIEVEMENT

- TIVE: A toolbox for identifying video instance segmentation errors Neurocomputing
  We designed a video instance segmentation evaluation framework called TIVE, which defines independent error types and weighs the impact of each type on mAP to differentiate model characteristics. By decomposing the quality metrics of segmentation results in the spatial and temporal dimensions, this framework can reveal potential shortcomings of models in spatial segmentation and temporal association.
- Weakly-supervised Video Instance Segmentation with Assignment-objective and Motion-adaptive Temporal Consistency TCE
  We propose a weakly supervised video instance segmentation method based on box supervision. It consists of two parts: for spatial supervision, we use bounding box projection and pixel similarity assumptions to calculate the loss; for temporal supervision, we employ pixel region feature matching to find temporally correlated pixel regions and compute similarity loss. This approach achieves accuracy close to that of fully supervised methods.
- MESI: A Dataset for Multimodal News Excerpt Segmentation and Identification TETCI
  We propose a new algorithm for long temporal video instance segmentation tasks, utilizing a coarse-to-fine multimodal fusion and hierarchical classification approach to achieve temporal segmentation and recognition of long videos. This method focuses on addressing the challenges of processing long videos and the inaccuracies in boundary segmentation and segment recognition. Additionally, a new benchmark has been constructed to evaluate the effectiveness of temporal segmentation in long videos.
- Decoupling Context from Computation: Plug-and-Play Modular Context Compression for Large Language Models In Progress
  We propose a novel framework for modular context compression in Large Language Models (LLMs). By leveraging an auxiliary neural branch—utilizing MLP or Attention mechanisms—we compress extensive textual context into a highly condensed set of reusable prefix tokens. These generated tokens function as plug-and-play modules that can be seamlessly prepended to new queries, effectively replacing the original lengthy context.

WORK EXPERIENCE

2025.04-Now Tiktok Algorithm Engineer, TikTok Search

Project Overview:

1. Optimizing and scaling LLMs and VLMs with task-specific capabilities for downstream tasks.
2. Built Intelligent/Automated Data Production Agent System for labeling(production) and evaluation.

Key Responsibilities: Algorithm Exploration, Design and Development

1. Search LLM(MLLM) Responsible for building multimodal generative recommendation (such as Onesug, etc.) and foundational NLP models for search, with a focus on participating in continued pretraining and various post-training tasks, exploring the limits of scalability in this scenario.
2. Intelligent Production To improve POI-related user metrics, we developed a multi-agent self-evolving framework. By extracting user preference experiences from behavioral data, the system generates optimized content/data to enhance online performance indicators.

2022.09-2025.04 Internships (at Kwai Technology Co., Ltd., JD.com, Inc., etc.) Algorithm Engineer

Project Overview:

1. In Kuaishou's live-streaming scenario, addressed various algorithmic requirements across business domains (including gaming, beauty, and lifestyle live streams), focusing on content understanding and description to provide algorithmic support for downstream recommendation systems.
2. In JD Logistics' scenario, designed and implemented an algorithm system to detect and identify non-compliant or violent handling behaviors during transportation, sorting, and terminal processing to effectively reduce cargo damage and enable clear accountability.
3. Participated in the exploration and optimization of cutting-edge 3D algorithms and contributed to building an AI Creation Platform, providing scene reconstruction and content generation algorithmic support for the company's 3D holographic interactive products (e.g., Naked-eye 3D frames, 3D video, and gaming platforms).

Key Responsibilities: Algorithm Exploration, Design and Development

1. Multimodal Live Stream LLM Research Participated in the development and construction of multimodal large model algorithms for understanding live streaming, including multi-task embedding extraction, query generation and system construction. Improved accuracy on hard samples by over 60%, resulting in a more robust system.
2. Multimodal Retrieval Solution Explored the possibility of Zero Shot in the context of continuously emerging game IPs, developing retrieval solutions that include visual features, multimodal features, and various retrieval methods, and conducted extensive experiments to validate these solutions. Design **multimodal large models** to extract more distinctive game content features, providing better feature references for retrieval.
3. Multimodal Logistic LLM Research Contributed to the violent sorting project, conducting detailed research and resource optimization (e.g., token compression) based on mainstream video multimodal large models. Built a high-quality logistics dataset and explored fine-tuning strategies, achieving a final Precision of 93.5% and Recall of 95.6%. Co-designed the violent behavior perception and recognition algorithm to establish a comprehensive detection system.
4. Naked-Eye 3D and Virtual Interaction Algorithms Researched the display principles of naked-eye 3D monitors. Designed and developed algorithms for generating and reconstructing naked-eye 3D display content, optimizing and expanding capabilities across multiple scenarios. This included novel view synthesis based on NeRF and 3DGS, and converting single-image 2D content to 3D content based on MPI. Expanded platform reach using Unity Shaders and JavaScript (e.g., Babylon.js) for 3D rendering of games and virtual characters. Developed Agent-based capabilities (action detection/generation) to create fully interactive AI virtual human characters.

2023.09-Now Large Language Models Research

1. Exploration based on multimodal large models of the LLAVA series. In terms of **shallow alignment**, research focused on components such as the Visual Encoder and cross-modal fusion. This included experiments on supporting high-resolution images, the capabilities of the Visual Encoder under different pre-training conditions, and the scalability of the Visual Encoder. These experiments verified the effectiveness of the algorithms. For **deep alignment,** more thorough methods of integrating visual and linguistic knowledge were explored, such as visual token compression, adaptive visual token extraction based on different inputs (visual knowledge selection), end-to-end visual tokenizers, and more efficient visual pre-training methods. Numerous experiments were conducted, confirming the validity of many hypotheses.
2. I have a strong research interest in Online Autonomous Continual Learning (ACL) for models, focusing on two primary directions: Model Optimization and Autonomous Interaction with the World. For model optimization, I investigate methods for partial parameter optimization (only optimizing necessary parameters) within a model. For example, transforming a model into an extremely sparse Mixture of Experts (MoE) model, where each parameter potentially represents a unique "expert." This approach is designed to effectively mitigate catastrophic forgetting and facilitate incremental learning. For autonomous interaction with the world, I aim to simulate the way humans interact with their environment. The goal is to allow the model to autonomously interact with the world, receive feedback during this process, and use that feedback to iteratively improve its own capabilities (self-correction and evolution).

SCHOOL EXPERIENCE

2021.09	Appointed to the School of Artificial Intelligence, BUPT
2022.12/2023.12	Graduate School First-Class Scholarship
2024.11	National Postgraduate Scholarship
Now	Reviewer for IEEE TCSVT and other top-tier journals

PROFESSIONAL SKILLS

English level： CET-4/6 with strong listening, speaking, reading, writing skills; quick at reviewing English texts；

Software skills： Proficient in Python, C; adept with Pytorch and Microsoft Office, etc.

Interests： Enjoys fitness, novels, anime, board games. Positive and optimistic with good habits.

ABOUT ME

Sincere and easy-going with strong adaptability; self-disciplined and principled, open to learning from others, willing to embrace challenges and take responsibility; good at self-reflection, diligent and practical, committed to giving 100% in learning and research, efficiently solving problems and challenges.